

Subspace:

(i) A non-empty subset W of vector space V is said to be a subspace of V if W is itself a vector space under same operations given in V .

(ii) Let V be a vector space and nonempty subset $W \subseteq V$ is said to be a subspace of V if it satisfies following conditions.

$$u_1 + u_2 \in W ; \forall u_1, u_2 \in W$$

$$\alpha u \in W ; \forall u \in W, \alpha \in F$$

(iii) Let V be a vector space and a non-empty subset W of V is said to be a subspace of V iff

$$\alpha u_1 + \beta u_2 \in W, \forall u_1, u_2 \in W, \alpha, \beta \in F$$

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Exercise # 6.1

Question no 6:

Is W is subspace of $\mathbb{R}^3$

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$$

$$(i) W = \{(x, y, z) : x^2 + y^2 + z^2 = 0\}$$

Solution:

Let $w_1, w_2 \in W$, $\alpha, \beta \in F$

$$w_1 = (x_1, y_1, z_1) : x_1^2 + y_1^2 + z_1^2 = 0 \quad \text{--- (i)}$$

$$w_2 = (x_2, y_2, z_2) : x_2^2 + y_2^2 + z_2^2 = 0 \quad \text{--- (ii)}$$

we have to show that

$$\alpha w_1 + \beta w_2 \in W$$

$$\begin{aligned} \alpha w_1 + \beta w_2 &= \alpha(x_1, y_1, z_1) + \beta(x_2, y_2, z_2) \\ &= (\alpha x_1, \alpha y_1, \alpha z_1) + (\beta x_2, \beta y_2, \beta z_2) \\ &= (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2) \end{aligned}$$

Now

$$\begin{aligned} &= \alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2 \\ &= \alpha(x_1 + y_1 + z_1) + \beta(x_2 + y_2 + z_2) \end{aligned}$$

from (i) and (ii)

$$= \alpha(0) + \beta(0)$$

$$= 0 + 0$$

$$= 0$$

$$\Rightarrow \alpha w_1 + \beta w_2 \in W$$

Hence W is subspace of $\mathbb{R}^3$

Is W is subspace of $\mathbb{R}^3$

$$(ii) W = \{(x, y, z) : x \geq 0\}$$

Let $w_1, w_2 \in W$, $\alpha, \beta \in F$

$$w_1 = (x_1, y_1, z_1) : x_1 \geq 0$$

$$w_2 = (x_2, y_2, z_2) : x_2 \geq 0$$

We have to show that

$$\alpha w_1 + \beta w_2 \in W$$

$$\alpha w_1 + \beta w_2 = \alpha(x_1, y_1, z_1) + \beta(x_2, y_2, z_2) \\ = (\alpha x_1, \alpha y_1, \alpha z_1) + (\beta x_2, \beta y_2, \beta z_2)$$

$$\alpha w_1 + \beta w_2 = (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2) \\ = \alpha x_1 + \beta x_2 \neq 0$$

Because $\alpha, \beta \in \mathbb{R}$ may be negative

$$\Rightarrow \alpha w_1 + \beta w_2 \notin W$$

Hence W is not subspace of $\mathbb{R}^3$

(iv) IS W is subspace of $\mathbb{R}^3$

$$W = \{(x, y, z) : x, z, y \text{ are rational}\}$$

$$\text{let } w_1, w_2 \in W : \alpha, \beta \in F$$

$$w_1 = (x_1, y_1, z_1) : x_1, y_1, z_1 \text{ are rational}$$

$$w_2 = (x_2, y_2, z_2) : x_2, y_2, z_2 \text{ are rational}$$

we have to show that

$$\alpha w_1 + \beta w_2 \in W$$

$$\alpha w_1 + \beta w_2 = \alpha(x_1, y_1, z_1) + \beta(x_2, y_2, z_2) \\ = (\alpha x_1, \alpha y_1, \alpha z_1) + (\beta x_2, \beta y_2, \beta z_2)$$

$$\alpha w_1 + \beta w_2 = (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2) \notin W$$

Because $\alpha, \beta \in \mathbb{R}$, may be irrational

$\Rightarrow$ Rational $\times$ Irrational = irrational.

(v) IS W is subspace of $\mathbb{R}^3$

$$W = \{(x, 0, z) : x, y \in \mathbb{R}\}$$

Question no 7:

Let V be the vector space of all real valued functions defined on $\mathbb{R}$. State which of following function are all the subspace of V

- (i) The set of all even function
- (ii) The set of all differentiable function
- (iii) The set $W = \{f \mid f(x) = kf(-x), k \in \mathbb{R} \text{ fixed}\}$
- (iv) The set $W = \{f \in V : \int \text{function of } x \, dx = 0\}$

Solution:

(i) There $V = \{f : f \text{ is real valued fun in } \mathbb{R}\}$
and

$W = \{f : f \text{ is all even functions}\}$

Let $f, g \in W$, $a, b \in \mathbb{F}$

$$f(-x) = f(x)$$

$$\vee f(-x) = f(x) \text{ even}$$

$$g(-x) = g(x)$$

$$f(-x) = -f(x) \text{ odd}$$

We have to show that $af + bg \in W$

Now

$$\begin{aligned} (af + bg)(-x) &= af(-x) + bg(-x) \\ &= af(x) + bg(x) \\ &= (af)(x) + (bg)(x) \\ &= (af + bg)(x) \in W \end{aligned}$$

Hence $af + bg \in W$

$\Rightarrow W$ is subspace of V .

(ii) $W = \{f : f \text{ is differentiable function}\}$

Let $f, g \in W$, $a, b \in \mathbb{F}$

$\Rightarrow f', g'$ exist

We have to show that $af + bg \in W$

$$\begin{aligned} \text{As } (af + bg)' &= af'(x) + bg'(x) \\ &= af' + bg' \in W \end{aligned}$$

Let $f'(x)$ and $g'(x)$ exist

Hence $af + bg \in W$

Therefore W is subspace of V

(iii) $W = \{f(x); f(x) = kf(-x), k \in \mathbb{R} \text{ fixed}\}$

Let $f, g \in W$, $a, b \in \mathbb{R}$

$$f(x) = kf(-x) \quad \text{--- (i)}$$

$$g(x) = kg(-x) \quad \text{--- (ii)}$$

we have to show that

$$af + bg \in W$$

$$(af + bg)(x) = af(x) + bg(x)$$

from (i) and (ii)

$$= a(kf(-x)) + b(kg(-x))$$

$$= k(af(-x)) + k(bg(-x))$$

$$= k[(af + bg)(-x)] \in W$$

Hence $af + bg \in W$

W is subspace of V .

$$(iv) W = \{f \in V : \int f(x) dx = 0\}$$

Let $f, g \in W$

$$\int f(x) dx = 0, \quad \int g(x) dx = 0$$

we have to show

$$af + bg \in W$$

as

$$\int (af + bg)(x) dx = \int af(x) dx + \int bg(x) dx$$

$$= a \int f(x) dx + b \int g(x) dx$$

$$= a(0) + b(0)$$

$$\int (af + bg)(x) dx = 0 \in W$$

Hence $af + bg \in W$

Therefore W is subspace of V .

Homework

Question no 8

Let V be the vectorspace of real polynomial of degree $\leq n$ to gether with zero polynomial. Determine whether or not W is subspace of V , where W consist of all polynomial and zero polynomial

- (i) write integral coefficients and of degree $\leq n$
- (ii) of degree ≤ 3
- (iii) write only even power of x of degree $\leq n$

ii. let $P(x) = a_0 + a_1x + \dots + a_nx^n$ where $a_0, a_1, \dots, a_n \in \mathbb{Z}$.

Then $P(x) \in W$ if α is a function or an irrational number then $\alpha P(x) = \alpha a_0 + \alpha a_1x + \dots + \alpha a_nx^n \notin W$ since $\alpha a_0, \alpha a_1, \dots, \alpha a_n$ are not integers. Thus W is not closed under scalar multiplication and so W is not a subspace of V

(ii) $W = \{P(x) : \deg p(x) \leq 3\}$

if $p_1(x), p_2(x) \in W$, then $\deg p_1(x) \leq 3$

$\deg p_2(x) \leq 3$. If $a, b \in \mathbb{R}$, then

$\deg (ap_1(x) + bp_2(x)) \leq 3$

Hence $ap_1(x) + bp_2(x) \in W$

Thus W is a subspace of V

(iii) $W = \{P(x) : P(x) = b_0 + b_1x^2 + b_2x^4 + \dots + b_{2k}x^{2k} \mid 2k \leq n\}$

Suppose $u, v \in W$

let $u = c_0 + c_1x^2 + c_2x^4 + \dots + c_{2k}x^{2k}$

$v = d_0 + d_1x^2 + d_2x^4 + \dots + d_{2k}x^{2k}$

Where some coefficients may be zero

if $a, b \in \mathbb{R}$, then

$au + bv = (ac_0 + bd_0) + (ac_1 + bd_1)x^2 + (ac_2 + bd_2)x^4 + \dots + (ac_{2k} + bd_{2k})x^{2k}$

Thus W is a subspace of V .

$x^{2k} \in W$

* Linear Combination

Let V be a vector space over field F . Let $v_1, v_2, \dots, v_n \in V$.

For any vector $v \in V$

Such that

$v = a_1 v_1 + a_2 v_2 + \dots + a_n v_n, \quad \forall a_i \in F$ is said to be a linear combination $v_1, v_2, \dots, v_n$

Q. Express the vector $(1, -2, -8) \in \mathbb{R}^3$ in linear combination of $(3, 0, -2)$ and $(2, -1, 5)$

$$v = a_1 v_1 + a_2 v_2, \quad \forall a_i \in F$$

$$\begin{aligned} (1, -2, -8) &= a_1 (3, 0, -2) + a_2 (2, -1, 5) \quad \dots (i) \\ &= (3a_1, 0 - 2a_1) + (2a_2 - a_2, -5a_2) \end{aligned}$$

$$(1, -2, -8) = (3a_1 + 2a_2, -a_2 - 2a_1 - 5a_2)$$

Comparing

$$3a_1 + 2a_2 = 1 \quad \dots (i)$$

$$-a_2 = -2 \quad \dots (ii)$$

$$-2a_1 - 5a_2 = -8 \quad \dots (iii)$$

from (ii)

$$\boxed{a_2 = 2}$$

Put in (i)

$$3a_1 + 2(2) = 1$$

$$3a_1 + 4 = 1$$

$$3a_1 = -3$$

$$\boxed{a_1 = -1}$$

Exercise 6.1

Linear Span / Spanning Set:-

Let S be non-empty subset of a vector space V . The set of all linear combinations of elements of S is called linear span of S denoted by $\langle S \rangle$.

Question no 12:

Show that each of following set of vectors generates $\mathbb{R}^3$.

(i) $\{(1, 2, 3), (0, 1, 2), (0, 0, 1)\}$

Solution:

Let $v = (x, y, z) \in \mathbb{R}^3$

$$v = a_1 v_1 + a_2 v_2 + a_3 v_3$$

$$(x, y, z) = a_1(1, 2, 3) + a_2(0, 1, 2) + a_3(0, 0, 1)$$

$$= (1a_1, 2a_1, 3a_1) + (0a_2, 1a_2, 2a_2) + (0a_3, 0a_3, 1a_3)$$

$$(x, y, z) = (a_1 + 0 + 0, 2a_1 + a_2 + 0, 3a_1 + 2a_2 + a_3)$$

Comparing

$$a_1 + 0 + 0 = x \quad \text{--- (i)}$$

$$2a_1 + a_2 + 0 = y \quad \text{--- (ii)}$$

$$3a_1 + 2a_2 + a_3 = z \quad \text{--- (iii)}$$

from (i)

$$\boxed{a_1 = x}$$

Put in (ii)

$$2x + a_2 = y$$

$$\boxed{a_2 = y - 2x}$$

Put value of a_1 and a_2 in (iii)

$$3x + 2(y - 2x) + a_3 = z$$

$$3x + 2y - 4x + a_3 - z = 0$$

$$\boxed{a_3 = z + x - 2y}$$

(Spanning Test)

$$\begin{array}{c|ccc} & 1 & 0 & 0 \\ \hline & 2 & 1 & 0 \\ & 3 & 2 & 1 \end{array}$$

$$= 1 \begin{array}{c|cc} 1 & 0 & 0 \\ \hline 2 & 1 & 1 \end{array} + 0 + 0$$

$$= (1 - 0)$$

$$= 1 \neq 0 \text{ span.}$$

Put all value of a_1, a_2 and a_3 in (R)

$$(x, y, z) = x(1, 2, 3) + (y - 2x)(0, 1, 2) + (z + x - 2y)(0, 0, 1)$$

$$\begin{aligned} &= (x, 2x, 3x) + (0, (y - 2x), 2(y - 2x)) + (0, 0, (x - 2y + z)) \\ &= (x, 2x + (y - 2x), 3x + 2(y - 2x) + x - 2y + z) \\ &= (x, y, z) \end{aligned}$$

Hence given set of vector span $V = \{(x, y, z) \in \mathbb{R}^3\}$

Therefore given set of vector generate $\mathbb{R}^3$

$$(ii) \{(1, 1, 1), (0, 1, 1), (0, 1, -1)\}$$

Linearly dependent and linearly Independent:-

Let V be a vector space over field F . The vectors $v_1, v_2, v_3, \dots, v_m \in V$ are said to be linearly dependent over F if

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0, \forall a_i \in F$$

and not all a_i are zero

Let V be a vector space over field F . The vectors $v_1, v_2, \dots, v_m \in V$ are said to be linearly independent over F if

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0, a_i \in F \text{ and each } a_i = 0$$

Example:

Show that the vectors $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ are linear independent.

Solution:

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = \vec{0}$$

$$\alpha_1 (1, 0, 0) + \alpha_2 (0, 1, 0) + \alpha_3 (0, 0, 1) = (0, 0, 0)$$

$$(\alpha_1, 0, 0) + (0, \alpha_2, 0) + (0, 0, \alpha_3) = (0, 0, 0)$$

$$(\alpha_1 + 0 + 0, 0 + \alpha_2 + 0, 0 + 0 + \alpha_3) = (0, 0, 0)$$

Comparing

$$\alpha_1 = 0 \quad \text{--- (i)}$$

$$\alpha_2 = 0 \quad \text{--- (ii)}$$

$$\alpha_3 = 0 \quad \text{--- (iii)}$$

$$\forall \alpha_i = 0$$

$\Rightarrow$ given vectors are linearly Independent

Exercise no 6.2

Question no 1:

Determine whether of the following vectors to $\mathbb{R}^3$ are linearly dependent or linearly independent

ii) $(1, -2, 4, 1), (2, 1, 0, -3), (1, -6, 1, 4)$

Solution:

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = \vec{0} \quad a_i \in F$$

$$a_1(1, -2, 4, 1) + a_2(2, 1, 0, -3) + a_3(1, -6, 1, 4) = (0, 0, 0, 0)$$

$$(a_1, -2a_1, 4a_1, a_1) + (2a_2, a_2, 0, -3a_2) + (a_3, -6a_3, a_3, 4a_3) = (0, 0, 0, 0)$$

$$(a_1 + 2a_2 + a_3, -2a_1 + a_2 + 0 - 6a_3, 4a_1 + 0 + a_3, a_1 - 3a_2 + 4a_3) = (0, 0, 0, 0)$$

$$(a_1 + 2a_2 + a_3, -2a_1 + a_2 - 6a_3, 4a_1 + 0 + a_3, a_1 - 3a_2 + 4a_3) = (0, 0, 0, 0)$$

Comparing

$$a_1 + 2a_2 + a_3 = 0 \quad \text{(i)}$$

$$-2a_1 + a_2 - 6a_3 = 0 \quad \text{(ii)}$$

$$4a_1 + 0 + a_3 = 0 \quad \text{(iii)}$$

$$a_1 - 3a_2 + 4a_3 = 0 \quad \text{(iv)}$$

Solving eq (i) and (iii)

$$a_1 + 2a_2 + a_3 = 0$$

$$-2a_1 + a_2 - 6a_3 = 0$$

$$\frac{a_1}{2} = \frac{-a_2}{-2} = \frac{a_3}{-2} = k$$

$$\left| \begin{array}{cc|cc} 2 & 1 & 1 & 1 \\ 1 & -6 & -2 & -6 \end{array} \right|$$

$$\frac{a_1}{-13} = \frac{a_2}{-4} = \frac{a_3}{5} = K$$

$$\frac{a_1}{-13} = K; \quad \frac{a_2}{-4} = K; \quad \frac{a_3}{5} = K$$

$$a_1 = -13K; \quad a_2 = -4K; \quad a_3 = 5K$$

Put in (iii) and (iv)

$$4(-13K) + 0 + 5K = 0$$

$$-52K + 5K = 0$$

Now put in (iv)

$$-8K - 3(-4K) + 4(5K) = 0$$

$$-8K + 12K + 20K = 0$$

Evaluation (iii) and (iv) are not satisfied only when $K=0$

$$a_1 = 0; \quad a_2 = 0; \quad a_3 = 0$$

$$\forall a_i = 0$$

given Set of vectors are linearly Independence.

$$(i) (1, 3, -1, 4), (3, 8, -5, 7), (2, 9, 4, 23)$$

Solution:

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = \vec{0}, \quad a_i \in F$$

$$a_1(1, 3, -1, 4) + a_2(3, 8, -5, 7) + a_3(2, 9, 4, 23)$$

$$(a_1, 3a_1, -a_1, 4a_1) + (3a_2, 8a_2, -5a_2, 7a_2) + (2a_3, 9a_3, 4a_3, 23a_3) = (0, 0, 0, 0)$$

$$a_1 + 3a_2 + 2a_3 = 0 \quad \text{--- (i)}$$

$$3a_1 + 8a_2 + 9a_3 = 0 \quad \text{--- (ii)}$$

$$-a_1 - 5a_2 + 4a_3 = 0 \quad \text{--- (iii)}$$

$$4a_1 - 7a_2 + 23a_3 = 0 \quad \text{--- (iv)}$$

Solving (i) and (ii)

$$\frac{a_1}{27-16} = \frac{-a_2}{9-6} = \frac{a_3}{8-9} = K$$

$$\frac{a_1}{11} = K ; \frac{-a_2}{3} = K ; \frac{a_3}{-1} = K$$

$$a_1 = 11K ; -a_2 = 3K ; a_3 = -1K$$

$$a_1 = 11K ; a_2 = -3K ; a_3 = -1K$$

Put in (iii) and (iv)

$$-11K - 5 - (3K) + 4(K) = 0$$

$$= -4K - 15K - 4K = 0$$

$$= 0 = 0$$

Satisfied

Now put in (iv)

$$4(11K) - 7(3K) + 23(-K) = 0$$

$$44K - 21K - 23K = 0$$

$$0 = 0$$

Satisfied

Eq (4) and (3) are Satisfied

Therefore given vectors are linearly Independence

Question no 2

Let $V = B(x)$ be the vector space of all polynomial of degree < 3 over F with given polynomial. Determine $u, v, w \in V$ are linearly depend and linearly independ.

$$(i) u = x^3 - 4x^2 + 2x + 3$$

$$v = x^3 + 2x^2 + 4x - 1$$

$$w = 2x^3 - x^2 + 3x + 3$$

Solution:

$$au + bv + cw = \overline{0}, \quad \forall a, b, c \in F$$

$$a(x^3 - 4x^2 + 2x + 3) + b(x^3 + 2x^2 + 4x - 1) + c(2x^3 - x^2 + 3x + 3) = 0x^3 + 0x^2 + 0x + 0$$

$$(ax^3 - 4ax^2 + 2ax + 3a) + (bx^3 + 2bx^2 + 4bx - b) + (2cx^3 - cx^2 + 3cx + 3c) = (0x^3 + 0x^2 + 0x + 0)$$

$$(ax^3 + bx^3 + 2cx^3) + (-4ax^2 + 2bx^2 - cx^2) + (2ax + 4bx + 3cx) + (3a - b + 3c) = 0$$

$$(a+b+2c)x^3 + (-4a+2b-c)x^2 + (2a+4b+3c)x + (3a-b+3c) = (0x^3 + 0x^2 + 0x + 0)$$

comparing coefficient of x^3, x^2, x

$$a + b + 2c = 0 \quad (i)$$

$$-4a + 2b - c = 0 \quad (ii)$$

$$2a + 4b + 3c = 0 \quad (iii)$$

$$3a - b + 3c = 0 \quad (iv)$$

Question No 5:

Suppose u, v and w are linearly independence vectors, Prove that:

(i) $u+v-2w, u-v-w, u+w$ are linearly independent

(ii) $u+v-3w, u+3v-w, u+w$ are linearly independent.

Question no 6:

Determine K so that the vectors $(1, -1, K-1)$, $(2, K, -4)$, $(0, 2+K, -8)$ in $\mathbb{R}^3$ are linearly dependent

Solution:

Since vector are linearly dependent

$$(1, -1, K-1) = a(2, K, -4) + b(0, 2+K, -8)$$

$$(1, -1, K-1) = (2a, Ka, -4a) + (0, (2+K)b, -8b)$$

$$(1, -1, K-1) = (2a + 0, Ka + (2+K)b, -4a - 8b)$$

Comparing

$$2a + 0 = 1 \quad \text{--- (i)}$$

$$Ka + (2b + bK) = -1 \quad \text{--- (ii)}$$

$$-4a - 8b = K-1 \quad \text{--- (iii)}$$

From (i)

$$a = \frac{1}{2}$$

Put $a = \frac{1}{2}$ in (3)

$$-4\left(\frac{1}{2}\right) - 8b = K-1$$

$$-2 - 8b = K-1$$

$$-8b = K+1$$

$$b = \frac{-(K+1)}{8}$$

Put values of a and b in 2

$$K\left(\frac{1}{2}\right) + (2+K)\left(\frac{K+1}{8}\right) = -1$$

x 8

$$4K + (2+K)(-K-1) = -8$$

$$4K + (-2K - 2 - K^2 - K) = -8$$

$$4K - 3K - 2 - K^2 = -8$$

$$-K^2 + K - 2 = -8$$

$$-(K^2 - K + 2) = -8$$

$$K^2 - K + 2 = 8$$

$$K^2 - K + 2 - 8 = 0$$

$$K^2 - K - 6 = 0$$

$$K^2 - 3K + 2K - 6 = 0$$

$$K(K-3) + 2(K-3) = 0$$

$$(K+2)(K-3) = 0$$

$$K+2=0 \quad ; \quad K-3=0$$

$$\boxed{K = -2}$$

$$\boxed{K = 3}$$

Basis:

A set of linearly independent vectors spanning the vector space V is called Basis of V

Example:

(i) $\{(1,0), (0,1)\}$

(ii) $\{(1,0,0), (0,1,0), (0,0,1)\}$

(iii) $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$

Example:-

Show that vectors $(1,0), (0,1)$ are form basis for $\mathbb{R}^2$

Solution:

Find we need to check vectors are linearly independent

$$a v_1 + b v_2 = 0 \quad \forall a, b \in F$$

$$a(1,0) + b(0,1) = (0,0)$$

~~$(a+b, b)$~~

$$(a+0, 0+b) = (0, 0)$$

$$a=0 \quad ; \quad b=0$$

$$\boxed{a=b=0}$$

given vectors are linearly independent

Now check span.

$$= \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$= 1 - 0$$

$\boxed{= 1 \neq 0}$ by spanning test we concluded that

given set of vector span $\mathbb{R}^2$ Therefore vectors are from the basis of $\mathbb{R}^2$.

Question no 9:

Determine whether or not given set of vectors is a basis for $\mathbb{R}^2$

$$(i) \{ (1, 1), (3, 1) \}$$

$$(ii) \{ (2, 1), (1, -1) \}$$

Question no 10:

Determine whether or not given set of vectors is a basis for $\mathbb{R}^3$

(i) $\{(1, 2, -1), (0, 3, 1), (1, -5, 3)\}$

(ii) $\{(2, 4, -3), (0, 1, 1), (0, 1, -1)\}$

Question no 9 ii) Solution:-

(i) $\{(1, 1), (3, 1)\}$

$$av_1 + bv_2 = \vec{0}, \forall a, b \in F$$

$$a(1, 1) + b(3, 1) = (0, 0)$$

$$a + 3b = 0 \quad \text{--- (i)}$$

$$a + b = 0 \quad \text{--- (ii)}$$

Subtract (2) from (1)

$$a + 3b = 0$$

$$-a + b = 0$$

$$2b = 0$$

$$b = 0$$

$$\boxed{b = 0}$$

Put in (2)

$$a + 0 = 0$$

$$\boxed{a = 0}$$

Hence $\boxed{a = b = 0}$

Vectors are linearly independent. Now check span

$$\begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix}$$

$$= 1 - 3$$

$$= -2 \neq 0$$

By spanning test given vectors span $\mathbb{R}^2$ given
set of therefore $\wedge$ vectors span $\mathbb{R}^2$.

Exercise no 6.3

Question no 1:-

Find basis and dimension of subspace W of $\mathbb{R}^3$.

(i) $(1, 4, -1, -3), (2, 1, -3, -1), (0, 2, 1, -5)$

Solution:-

Find we check vectors are linearly independent.

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = \vec{0} \quad \alpha_i \in F$$

$$\alpha_1(1, 4, -1, -3) + \alpha_2(2, 1, -3, -1) + \alpha_3(0, 2, 1, -5) = (0, 0, 0, 0)$$

$$\alpha_1 + 2\alpha_2 + 0 = 0 \quad \text{--- (i)}$$

$$4\alpha_1 + \alpha_2 + 2\alpha_3 = 0 \quad \text{--- (ii)}$$

$$-\alpha_1 - 3\alpha_2 + \alpha_3 = 0 \quad \text{--- (iii)}$$

$$3\alpha_1 - \alpha_2 - 5\alpha_3 = 0 \quad \text{--- (iv)}$$

Solving (i) and (iii)

$$\frac{\alpha_1}{4-0} = \frac{-\alpha_2}{2-0} = \frac{\alpha_3}{1-8} = K$$

$$\frac{\alpha_1}{4} = K, \quad \frac{-\alpha_2}{2} = K, \quad \frac{\alpha_3}{-7} = K$$

$$\boxed{\alpha_1 = 4K}, \quad \boxed{\alpha_2 = -2K}, \quad \boxed{\alpha_3 = -7K}$$

Put values of α_i in eq (ii) and (iv)

First Put in (3)

$$-4(K-3) - (-2K) - 7K = 0$$

$$-4K + 6K - 7K = 0 \quad \boxed{\text{not satisfied}}$$

Put in (4)

$$3(4K) + 2K + 35K = 0$$

$$\boxed{\text{not satisfied}}$$

eq (iii) and (iv) are not satisfied

Satisfied $K=0$

$$\alpha_1 = 0, \alpha_2 = 0, \alpha_3 = 0$$

$$\forall \alpha_1 = 0$$

Given set of vector are linearly independent

Now by reference of theorem. The independent set of vector span $\mathbb{R}^3$

So

$$\{(1, 4, -1, 3), (2, 1, -3, -1), (0, 2, 1, -5)\}$$

is set of Basis dimensions = 3

Question no 2:

Find Basis and dimensional of subspace w of $\mathbb{R}^3$

$$i) (1, -4, -2, 1), (1, -3, -1, 2), (3, -8, -2, 7)$$

Solutions:-

Find we check vectors are linearly independent

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0$$

$$\alpha_1 (1, -4, -2, 1) + \alpha_2 (1, -3, -1, 2) + \alpha_3 (3, -8, -2, 7) = (0, 0, 0, 0)$$

$$\alpha_1 + \alpha_2 + 3\alpha_3 = 0 \quad \text{--- (1)}$$

$$-4\alpha_1 - 3\alpha_2 - 8\alpha_3 = 0 \quad \text{--- (2)}$$

$$-2\alpha_1 - 3\alpha_2 - 2\alpha_3 = 0 \quad \text{--- (3)}$$

$$\alpha_1 + 2\alpha_2 + 7\alpha_3 = 0 \quad \text{--- (4)}$$

Solving (1) and (2)

$$\frac{\alpha_1}{-8+9} = \frac{-\alpha_2}{-8+12} = \frac{\alpha_3}{-3+4} = k$$

$$\frac{\alpha_1}{1} = k, \quad \frac{-\alpha_2}{4} = k, \quad \frac{\alpha_3}{-3+4} = k$$

$$\frac{\alpha_1}{1} = k, \quad \frac{-\alpha_2}{4} = k, \quad \frac{\alpha_3}{1} = k$$

$$\alpha_1 = K \quad -\alpha_2 = 4K \quad \alpha_3 = K$$

$$\boxed{\alpha_1 = K} \quad \boxed{\alpha_2 = -4K} \quad \boxed{\alpha_3 = K}$$

Put values in eq. (3) and eq. (4)

$$-2(K) - (-4K) - 2K = 0$$

$$-2K + 4K - 2K = 0$$

$$\boxed{0=0} \text{ Satisfied}$$

$$K + 2(-4K) + 7(K) = 0$$

$$K - 8K + 7K = 0$$

$$\boxed{0=0} \text{ satisfied}$$

eq. (3) and (4) are satisfied

So given Set of vectors is linearly dependent
Hence the vector $(3, -8, -2, 7)$ can be written in combination of other two vectors.

Now check remaining vectors are linearly independent.

$$\alpha_1 v_1 + \alpha_2 v_2 = \vec{0}$$

$$\alpha_1 + \alpha_2 = 0 \quad \text{--- (i)}$$

$$-4\alpha_1 - 3\alpha_2 = 0 \quad \text{--- (ii)}$$

$$-2\alpha_1 - \alpha_2 = 0 \quad \text{--- (iii)}$$

$$\alpha_1 + 2\alpha_2 = 0 \quad \text{--- (iv)}$$

add eq. (i) and (iii)

$$\alpha_1 + \alpha_2 = 0$$

$$-2\alpha_1 - \alpha_2 = 0$$

$$\underline{-\alpha_1 = 0}$$

$$\alpha_1 = 0$$

Put in (i)

$$\alpha_3 = 0$$

vectors are linearly independent. So by reference of Theorem The vectors set of vectors also span R^3

Therefore $\{(1, -4, 2, 1), (1, -3, -1, 2)\}$
a set of Basis dimensional = 2

Linear Algebra

Inner Product space:

Let V be a vector space over field F of real or complex nos. A mapping $\langle \cdot, \cdot \rangle: V \times V \rightarrow F$ is said to be an inner product on V , if following conditions are satisfied

(i) $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ iff $v=0$, $\forall v \in V$

(ii) $\langle v_1, v_2 \rangle = \overline{\langle v_2, v_1 \rangle}$ $\forall v_1, v_2 \in V$

Here $\overline{\langle v_2, v_1 \rangle}$ is conjugate of $\langle v_2, v_1 \rangle$

(iii) $\langle av_1 + bv_2, v_3 \rangle = a\langle v_1, v_3 \rangle + b\langle v_2, v_3 \rangle$, $\forall v_1, v_2, v_3 \in V$

if F taken as of field of real no's only Then $\langle v_2, v_1 \rangle = \langle v_1, v_2 \rangle$ and so condition (ii) becomes $\langle v_1, v_2 \rangle = \langle v_2, v_1 \rangle$

The pair $(V, \langle \cdot, \cdot \rangle)$ is called an inner product space.

Exercise no 7.1

Question no 1:

Let $u=(u_1, u_2)$ and $v=(v_1, v_2)$ belong to $\mathbb{R}^2$

(i) verify $\langle u, v \rangle = u_1v_1 - 2u_1v_2 - 2u_2v_1 + 5u_2v_2$ is an inner product on $\mathbb{R}^2$

(ii) For what value of K

$\langle u, v \rangle = u_1v_1 - 3u_1v_2 - 3u_2v_1 + Ku_2v_2$ is an inner product on $\mathbb{R}^2$

Solution:

$u=(u_1, u_2)$, $v=(v_1, v_2)$ define as

$$\langle u, v \rangle = u_1v_1 - 2u_1v_2 - 2u_2v_1 + 5u_2v_2$$

(i) $\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0$ iff $u=0$ $\forall u \in V$

$$\langle u, u \rangle = u_1u_1 - 2u_1u_2 - 2u_2u_1 + 5u_2u_2 \Rightarrow \langle u, u \rangle = u_1^2 - 4u_1u_2 + 5u_2^2$$

$$\langle u, u \rangle = u_1^2 - 4u_1u_2 + (2u_2)^2 = (2u_2)^2 + 5u_2^2$$

$$= (u_1 - 2u_2)^2 - 4u_2^2 + 5u_2^2$$

$$\langle u, u \rangle = (u_1 - 2u_2)^2 + u_2^2 \geq 0$$

and

$$\langle u, u \rangle = 0$$

$$\Leftrightarrow (u_1 - 2u_2)^2 + u_2^2 = 0$$

$$\Leftrightarrow (u_1 - 2u_2)^2 = 0 \quad ; \quad u_2^2 = 0$$

$$\Leftrightarrow u_1 - 2u_2 = 0 \quad ; \quad u_2 = 0$$

$$\Leftrightarrow u_1 - 2(0) = 0 \quad ; \quad u_2 = 0$$

$$\Leftrightarrow u_1 = 0 \quad ; \quad u_2 = 0$$

$$\Leftrightarrow u = 0$$

$$\therefore u = (u_1, u_2) = (0, 0)$$

$$u = 0$$

$$(iii) \langle u, v \rangle = \langle v, u \rangle$$

L.H.S

$$\langle u, v \rangle = u_1 v_1 - 2u_1 v_2 - 2u_2 v_1 + 5u_2 v_2$$

$$= v_1 u_1 - 2v_2 u_1 - 2v_1 u_2 + 5v_2 u_2$$

$$= v_1 u_1 - 2v_1 u_2 - 2v_2 u_1 + 5v_2 u_2$$

$$\therefore \langle u, v \rangle = \langle v, u \rangle$$

$$(iv) \langle au + bv, w \rangle = a \langle u, w \rangle + b \langle v, w \rangle, \quad u, v, w \in \mathbb{R}^2$$

$$au + bv = a(u_1, u_2) + b(v_1, v_2) = (au_1, au_2) + (bv_1, bv_2)$$

$$au + bv = \underbrace{(au_1 + bv_1)}_{u_1}, \underbrace{(au_2 + bv_2)}_{u_2}, \quad w = \underbrace{(w_1)}_{v_1}, \underbrace{(w_2)}_{v_2}$$

L.H.S

$$\langle au + bv, w \rangle = (au_1 + bv_1)w_1 - 2(au_1 + bv_1)w_2 - 2(au_2 + bv_2)w_1 + 5(au_2 + bv_2)w_2$$

$$= au_1 w_1 + bv_1 w_1 - 2au_1 w_2 - 2bv_1 w_2 - 2au_2 w_1 - 2bv_2 w_1 + 5au_2 w_2 + 5bv_2 w_2$$

$$= \langle au + bv, w \rangle = a(u_1 w_1 - 2u_1 w_2 - 2u_2 w_1 + 5u_2 w_2) + b(v_1 w_1 - 2v_1 w_2 - 2v_2 w_1 + 5v_2 w_2)$$

$$\langle au + bv, w \rangle = a \langle u, w \rangle + b \langle v, w \rangle$$

all properties are satisfied so $\langle u, v \rangle$ is an inner product on $\mathbb{R}^2$

Linear Algebra

Question no 5:

Let V be the vector space $P(x)$ of all real valued polynomial over $\mathbb{R}$. Show that

$$\langle f, g \rangle = \int_0^1 f(t) g(t) dt \text{ defines inner product on } \mathbb{R}^2$$

Solution:

$$\bullet \langle f, f \rangle \geq 0 \text{ and } \langle f, f \rangle = 0 \text{ iff } f=0, \forall f \in V$$

$$\langle f, f \rangle = \int_0^1 f(t) f(t) dt$$

$$\langle f, f \rangle = \int_0^1 [f(t)]^2 dt \geq 0$$

and

$$\langle f, f \rangle = 0$$

$$\Leftrightarrow \int_0^1 [f(t)]^2 dt = 0$$

$$\Leftrightarrow \int_0^1 f(t) dt = 0$$

$$\Leftrightarrow f(t) = 0$$

$$\Leftrightarrow f = 0$$

$$\bullet \langle f, g \rangle = \langle g, f \rangle$$

$$\langle f, g \rangle = \int_0^1 f(t) g(t) dt$$

$$= \int_0^1 g(t) f(t) dt$$

$$\langle f, g \rangle = \langle g, f \rangle$$

Let $h \in V$

$$3) \langle af+bg, h \rangle = a\langle f, h \rangle + b\langle g, h \rangle$$

$$\langle (af+bg), h \rangle = \int_0^1 (af+bg)(t) h(t) dt$$

$$= \int_0^1 af(t)h(t) dt + \int_0^1 bg(t)h(t) dt$$

$$= a \int_0^1 f(t)h(t) dt + b \int_0^1 g(t)h(t) dt$$

$$\langle (af+bg), h \rangle = a\langle f, h \rangle + b\langle g, h \rangle$$

all axioms are satisfied Hence $\langle \cdot, \cdot \rangle$ is inner product on V .

Example no 4:-

Let V be the vector space of all real valued continuous function on the interval $a \leq t \leq b$. Then $f, g \in V$

$$\langle f, g \rangle = \int_a^b f(t)g(t)dt \text{ is an inner Product on } V.$$

Example no 3:-

Let $u, v \in \mathbb{R}^2$, $u = (x_1, x_2)$, $v = (y_1, y_2)$ Then

$\langle u, v \rangle = x_1y_1 - x_1y_2 - x_2y_1 + 3x_2y_2$ is an inner Product on $\mathbb{R}^2$

Solution:-

$$(i) \langle u, u \rangle \geq 0 \text{ and } \langle u, u \rangle = 0 \text{ iff } u=0, \forall u \in \mathbb{R}^2$$

$$\langle u, u \rangle = x_1x_1 - x_1x_2 - x_2x_1 + 3x_2x_2 \Rightarrow x_1^2 - 2x_1x_2 + 3x_2^2$$

$$= x_1^2 - 2x_1x_2 + (x_2)^2 - (x_2)^2 + 3x_2^2$$

$$\langle u, u \rangle = (x_1 - x_2)^2 + 2x^2 \geq 0$$

$$\text{and } \langle u, u \rangle = 0$$

$$\Leftrightarrow (x_1 - x_2)^2 + 2x^2 = 0$$

$$\Leftrightarrow (x_1 - x_2)^2 = 0, x_2^2 = 0$$

$$\Leftrightarrow x_1 - x_2 = 0, x_2 = 0$$

$$\Leftrightarrow x_1 = 0, x_2 = 0 \quad \checkmark$$

$$\Leftrightarrow u = 0$$

condition (i) is true

$$(ii) \langle u, v \rangle = \langle v, u \rangle$$

$$\langle u, v \rangle = x_1 y_1 - x_1 x_2 - x_2 x_1 + 3x_2 y_2$$

$$= y_1 x_1 - x_2 x_1 - x_1 x_2 + 3y_2 x_2$$

$$= y_1 x_2 - x_1 x_2 - x_2 x_1 + 3y_2 x_2$$

$\langle u, v \rangle = \langle v, u \rangle$ is true

$$(iii) \langle au + bv, w \rangle = a \langle u, w \rangle + b \langle v, w \rangle$$

$$au + bv = a(x_1, x_2) + b(y_1, y_2)$$

$$= (ax_1, ax_2) + (by_1, by_2)$$

$$au + bv = (ax_1 + by_1, ax_2 + by_2), w(z_1, z_2)$$

$$\langle au + bv, w \rangle = (ax_1 + by_1)z_1 - (ax_1 + by_1)z_2 - (ax_2 + by_2)z_1 + 3(ax_2 + by_2)z_2$$

$$= ax_1 z_1 + by_1 z_1 - ax_1 z_2 - by_1 z_2 - ax_2 z_1 - by_2 z_1 + 3ax_2 z_2 + 3by_2 z_2$$

$$= a(x_1 z_1 - x_1 z_2 - x_2 z_1 + 3x_2 z_2) + b(y_1 z_1 - y_1 z_2 - y_2 z_1 + 3y_2 z_2)$$

$$= a \langle u, w \rangle + b \langle v, w \rangle$$

all axioms conditions are true satisfied So $\langle \cdot, \cdot \rangle$ is an inner product on V .

The vectors u and v are said to be orthogonal iff $\theta = 90^\circ$, in which $\angle u, v = 0$, So u and v are orthogonal iff $\angle u, v = 0$
 $\frac{1}{\|u\|} \cdot \frac{1}{\|v\|}$

if u is orthogonal to v then we write $u \perp v$ and read " u is orthogonal to v ".

Example: Let $x = (1, -1, 2)$ and $y = (-1, 1, +1) \in \mathbb{R}^3$.
Then Show that $x \perp y$ (x is orthogonal to y)
Solution:

$$\begin{aligned}\angle x, y &= x_1 x_2 + y_1 y_2 + z_1 z_2 \\ &= (1)(-1) + (-1)(1) + (2)(1) \\ &= -1 - 1 + 2 \\ &= -2 + 2\end{aligned}$$

$$\angle x, y = 0$$

Hence $x \perp y$

20-May, 2025

Linear Algebra

Tuesday

Norm / length of vector:-

Let V be an inner product space and $u \in V$. Then the real no. $\sqrt{\langle u, u \rangle}$ is called norm of u and is denoted by $\|u\|$ if $\|u\| = 1$ i.e. if $\langle u, u \rangle = 1$, then u is called unit norm or normalized vector.

$$\|u\| = \sqrt{\langle u, u \rangle}$$

Example:- Find Norm of $(3, 5) \in \mathbb{R}^2$. Also Find Normalize vector.

$$\text{Let } u = (3, 5)$$

$$\begin{aligned} \langle u, u \rangle &= 3^2 + 5^2 \\ &= 9 + 25 \\ &= 34 \end{aligned}$$

$$\begin{aligned} \|u\| &= \sqrt{\langle u, u \rangle} \\ &= \sqrt{34} \end{aligned}$$

$$\hat{u} = \frac{u}{\|u\|}$$

Normalize vector.

$$= \frac{(3, 5)}{\sqrt{34}}$$

$$= \left(\frac{3}{\sqrt{34}}, \frac{5}{\sqrt{34}} \right)$$

Exercise no 7.1

Question no 2: Find norm of $(2, 3) \in \mathbb{R}^2$ w.r.t(i) The Euclidean inner product in $\mathbb{R}^2$ (ii) The inner product define as $\langle u, v \rangle = u_1 v_1 - 2u_1 v_2 - 2u_2 v_1 + 5u_2 v_2$

Solution:-

$$\text{Let } u = (2, 3) \in \mathbb{R}^2$$

$$\begin{aligned} \|u\| &= \sqrt{\langle u, u \rangle} \\ &= \sqrt{2^2 + 3^2} \\ &= \sqrt{4 + 9} \\ &= \sqrt{13} \end{aligned}$$

$$\text{ii) } \|u\| = \langle u, u \rangle = \sqrt{u_1 u_1 - 2u_1 u_2 - 2u_2 u_1 + 5u_2 u_2}$$

$$= \sqrt{u_1^2 - 4u_1 u_2 + 5u_2^2}$$

$$= \sqrt{(2)^2 - 4(2)(3) + 5(3)^2}$$

$$= \sqrt{4 - 24 + 45}$$

$$= \sqrt{49 - 24}$$

$$\|u\| = \sqrt{25}$$

$$\|u\| = 5$$

Question no 3. Find Norm of $(\frac{1}{2}, \frac{1}{4}, \frac{1}{3}, \frac{1}{6}) \in \mathbb{R}^4$ w.r.t Euclidean inner product.

Solution

$$\text{Let } u = (\frac{1}{2}, \frac{1}{4}, \frac{1}{3}, \frac{1}{6})$$

$$\|u\| = \sqrt{(\frac{1}{2})^2 + (\frac{1}{4})^2 + (\frac{1}{3})^2 + (\frac{1}{6})^2}$$

$$= \sqrt{\frac{1}{4} + \frac{1}{16} + \frac{1}{9} + \frac{1}{36}}$$

Exercise no 7-1

Cauchy Schwarz Inequality :-

Let V be an inner product space and $u, v \in V$. The Cauchy Schwarz inequality is $|\langle u, v \rangle| \leq \|u\| \|v\|$

Question no 4

Let V denotes the vector space of 2×2 matrices over $\mathbb{R}$ if $A, B \in V$ and $\text{Tr}(A)$ (called Trace of A) denote the sum of main diagonal entries of A , show that $\langle A, B \rangle = \text{Tr}(B^t A)$ is an inner product on V . also find norm of $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$

Solution:

Since $A, B \in V$

$$A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}, \quad B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$$

$$B^t A = \begin{bmatrix} b_1 & b_3 \\ b_2 & b_4 \end{bmatrix} \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$$

$$\begin{bmatrix} a_1 b_1 + a_3 b_3 & a_2 b_1 + a_4 b_3 \\ a_1 b_2 + a_3 b_4 & a_2 b_2 + a_4 b_4 \end{bmatrix}$$

$$\langle A, B \rangle = \text{Tr}(B^t A) = a_1 b_1 + a_3 b_3 + a_2 b_2 + a_4 b_4$$

last part:-

$$\begin{aligned} \|A\| &= \sqrt{\langle A, A \rangle} = \sqrt{a_1^2 + a_2^2 + a_3^2 + a_4^2} \\ &= \sqrt{(1)^2 + (2)^2 + (3)^2 + (-4)^2} \\ &= \sqrt{1+4+9+16} \\ &= \sqrt{30} \end{aligned}$$

(i) $\langle A, A \rangle \geq 0$ and $\langle A, A \rangle = 0$ iff $A = 0$

$$\langle A, A \rangle = a_1^2 + a_2^2 + a_3^2 + a_4^2$$

$$\langle A, A \rangle = a_1^2 + a_2^2 + a_3^2 + a_4^2 \geq 0$$

and $\langle A, A \rangle = 0$

$$\Leftrightarrow a_1^2 + a_2^2 + a_3^2 + a_4^2 = 0$$

$$\Leftrightarrow a_1^2 = 0, a_2^2 = 0, a_3^2 = 0, a_4^2 = 0$$

$$\Leftrightarrow a_1 = 0, a_2 = 0, a_3 = 0, a_4 = 0$$

$$\Leftrightarrow A = 0$$

$$(i) \langle A, B \rangle = \langle B, A \rangle$$

L.H.S

$$\begin{aligned} \langle A, B \rangle &= a_1 b_1 + a_3 b_3 + a_2 b_2 + a_4 b_4 \\ &= b_1 a_1 + b_3 a_3 + b_2 a_2 + b_4 a_4 \end{aligned}$$

$$\langle A, B \rangle = \langle B, A \rangle$$

$$(ii) \langle \alpha A + \beta B, C \rangle = \alpha \langle A, C \rangle + \beta \langle B, C \rangle, \text{ for } C \in V$$

$$C = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix}$$

L.H.S

$$\langle \alpha A + \beta B, C \rangle = \text{Tr}(C^T (\alpha A + \beta B))$$

$$C^T (\alpha A + \beta B) = \begin{bmatrix} c_1 & c_3 \\ c_2 & c_4 \end{bmatrix} \left(\alpha \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} + \beta \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} \right)$$

$$= \begin{bmatrix} c_1 & c_3 \\ c_2 & c_4 \end{bmatrix} \begin{bmatrix} \alpha a_1 + \beta b_1 & \alpha a_2 + \beta b_2 \\ \alpha a_3 + \beta b_3 & \alpha a_4 + \beta b_4 \end{bmatrix}$$

$$C^T (\alpha A + \beta B) = \begin{bmatrix} c_1 (\alpha a_1 + \beta b_1) + c_3 (\alpha a_3 + \beta b_3) & c_1 (\alpha a_2 + \beta b_2) + c_3 (\alpha a_4 + \beta b_4) \\ c_2 (\alpha a_1 + \beta b_1) + c_4 (\alpha a_3 + \beta b_3) & c_2 (\alpha a_2 + \beta b_2) + c_4 (\alpha a_4 + \beta b_4) \end{bmatrix}$$

$$\langle \alpha A + \beta B, C \rangle = \text{Tr}(C^T (\alpha A + \beta B)) = c_1 (\alpha a_1 + \beta b_1) + c_3 (\alpha a_3 + \beta b_3) + c_2 (\alpha a_2 + \beta b_2) + c_4 (\alpha a_4 + \beta b_4)$$

$$\begin{aligned} &= \alpha a_1 c_1 + \beta b_1 c_1 + \alpha a_3 c_3 + \beta b_3 c_3 + \alpha a_2 c_2 + \beta b_2 c_2 + \alpha a_4 c_4 + \beta b_4 c_4 \\ &= \alpha (a_1 c_1 + a_3 c_3 + a_2 c_2 + a_4 c_4) + \beta (b_1 c_1 + b_3 c_3 + b_2 c_2 + b_4 c_4) \end{aligned}$$

$$\langle \alpha A + \beta B, C \rangle = \alpha \langle A, C \rangle + \beta \langle B, C \rangle$$

Above all properties of inner product are satisfied.

Hence $\langle A, B \rangle$ is an inner product on V .

ie $(V, \langle \cdot, \cdot \rangle)$ is an inner product space.

Eigen values and Eigen Vectors:-

if A is an $n \times n$ matrix over R then a scalar $\lambda \in R$ is called an eigen value of A if there exist nonzero vector $v \in R^n$ such that $Av = \lambda v$. In this case v is called an eigen vector of A .

$$Av = \lambda v$$

$$Av = \lambda I v$$

$$Av - \lambda I v = 0$$

$$(A - \lambda I) v = 0$$

$$A - \lambda I = 0$$

$$\det(A - \lambda I) = 0$$

22-May-2025

Linear Algebra

Thursday

Exercise no 7.3

Q# 2: Find the eigen values and eigen vectors of $\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$.

Solution:

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$$

$$P(\lambda) = |A - \lambda I| = 0$$

$$A - \lambda I = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 - \lambda & 1 \\ -2 & 4 - \lambda \end{bmatrix}$$

or

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{vmatrix}$$

$$= \cancel{4-\lambda} - \cancel{1}$$

$$= (4-\lambda)(1-\lambda) - (-2)(1)$$

$$|A - \lambda I| = 4 - 4\lambda - \lambda + \lambda^2 + 2$$

$$P(\lambda) = |A - \lambda I| = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\lambda^2 - 3\lambda - 2\lambda + 6 = 0$$

$$\lambda(\lambda-3) - 2(\lambda-3) = 0$$

$$(\lambda-3)(\lambda-2) = 0$$

$$\lambda = 2, 3$$

are required Eigen values.

Now eigen vector corresponding to $\lambda = 2$

$$Av = \lambda v$$

$$\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 2 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 + x_2 \\ -2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}$$

$$x_1 + x_2 = 2x_1 \quad \text{--- (i)}$$

$$-2x_1 + 4x_2 = 2x_2 \quad \text{--- (ii)}$$

from (i)

$$-x_1 + x_2 = 0 \quad \text{--- (iv)}$$

$$-2x_1 + 2x_2 = 0 \quad \text{--- (v)}$$

let $x_1 = 1$ Put in (iv)

$$-1 + x_2 = 0$$

$$\boxed{x_2 = 1}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is eigen vector corresponding to $\lambda = 2$

Now Find eigen vector corresponding to $\lambda = 3$

$$AV = \lambda V$$

$$\begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 3 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 + x_2 \\ -2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 3x_1 \\ 3x_2 \end{bmatrix}$$

$$x_1 + x_2 = 3x_1 \quad \text{--- (i)}$$

$$-2x_1 + 4x_2 = 3x_2 \quad \text{--- (ii)}$$

from (i)

$$-2x_1 + x_2 = 0 \quad \text{(iii)}$$

$$-2x_1 + x_2 = 0 \quad \text{(iv)}$$

let $x_1 = 1$

Put in (iii) we get $x_2 = 2$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

is eigen vector corresponding to $\lambda = 3$

$$(ii) \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Solution:

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{bmatrix}$$

$$|A - \lambda I| = -\lambda^3 + 7\lambda^2 - 11\lambda + 5$$

$$\text{Now } |A - \lambda I| = 0$$

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

$$\lambda = 1, \lambda^2 - 6\lambda + 5 = 0$$

$$\lambda = 1, 5$$

The eigen values are

$$\lambda = 1, 1, 5$$

$$Av = \lambda v$$

→ Thursday

Quiz 1

$$6 \cdot 1 + 6 \cdot 2$$

Quiz 2

$$6 \cdot 3 + 7 \cdot 1 + 7 \cdot 3$$

$$\begin{array}{c|cccc} 1 & 1 & -7 & 11 & -5 \\ & \downarrow & 1 & -6 & 5 \\ & 1 & -6 & 5 & 0 \end{array}$$

$$\lambda = 1, \lambda^2 - 6\lambda + 5 = 0$$

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 1 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Exercise no 7.3

Question no (1-6)